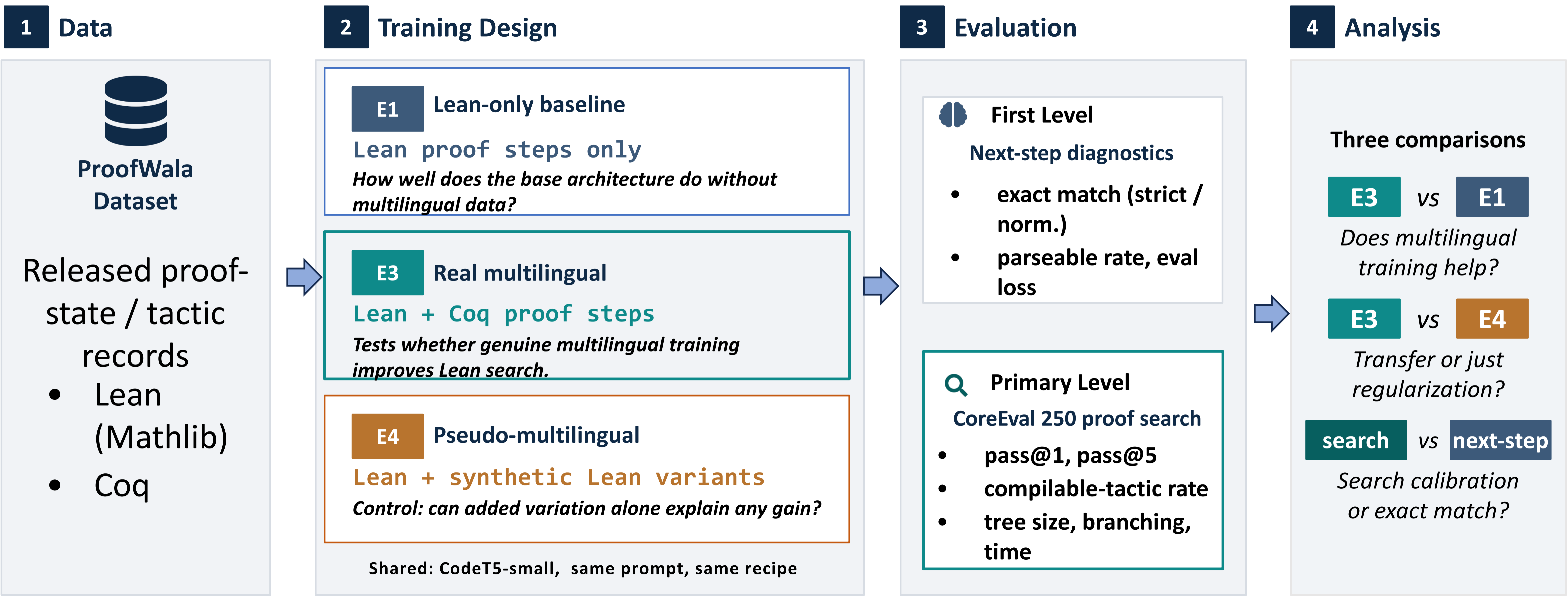
1 Data
ProofWala Dataset
Released proof-state / tactic records
• Lean (Mathlib)
• Coq
2 Training Design
E1 Lean-only baseline
Lean proof steps only
How well does the base architecture do without multilingual data?
E3 Real multilingual
Lean + Coq proof steps
Tests whether genuine multilingual training improves Lean search.
E4 Pseudo-multilingual
Lean + synthetic Lean variants
Control: can added variation alone explain any gain?
Shared: CodeT5-small, same prompt, same recipe
3 Evaluation
First Level
Next-step diagnostics
• exact match (strict / norm.)
• parseable rate, eval loss
Primary Level
CoreEval 250 proof search
• pass@1, pass@5
• compilable-tactic rate
• tree size, branching, time
4 Analysis
Three comparisons
E3 vs E1
Does multilingual training help?
E3 vs E4
Transfer or just regularization?
search vs next-step
Search calibration or exact match?